

Research Article

Intelligent Direct Torque Control Approach for BLDC Motorization in Electric Transportation Systems: A Review

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Abstract

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Electric vehicles (EVs) are a popular choice for future transportation because of their economy, sustainability, and zero emissions during operation. In-wheel drive has been one of the main research focus areas in the last decade. Because of their exceptional performance, torque-speed characteristics, high power-to-weight ratio, long service life, and noiseless operation, BLDC motors are used for in-wheel applications. This work proposes and simulates a direct torque control (DTC) switching technique for BLDC motors in EV propulsion systems using MATLAB/Simulink. Compared with conventional switching methods, the simulation results demonstrate efficient torque control and reduced torque ripple amplitude. Improvements in the torque controllability of the in-wheel motor lead to a safer and more efficient electric vehicle. The simulation results of the proposed switching scheme are satisfactory and show that the system operates as intended.

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I. INTRODUCTION

The decision to use electricity instead of fossil fuels for automobile propulsion is not new. Long-term efforts have been made by manufacturers and scientists to create and improve electric vehicles. In 1839, Robert Anderson built the first electric-powered vehicle. David Salomon created an electric vehicle with a light battery and motor in 1870. Performance suffered because of the weight of the batteries at that time [1]. Due to their higher mileage and the lack of adequate infrastructure for charging electric car batteries, hybrid vehicles are becoming more well known. In-wheel technology can improve electric vehicle performance and safety. By using in-wheel generation, an intelligent fully electronically controlled vehicle (IFECV) is produced by employing electric drive and intelligent control structures

rather than conventional hydraulic and pneumatic control methods [2]. Figure 1 shows the schematic diagram of an IFECV four-wheel drive system. Electric vehicle performance, controllability, and safety can be improved by enhancing the overall performance of the in-wheel motors and their controllers. In recent years, manufacturers have employed a large number of electric motors in vehicles. Induction, BLDC, brushed DC, and switched reluctance motors have all been compared, and BLDC motors are recommended for high-performance electric vehicles. Several sensorless control techniques, such as back-EMF estimation, flux-linkage-based approaches, and free-wheeling diode conduction, can be used to commutate BLDC motors in place of sensors [3].

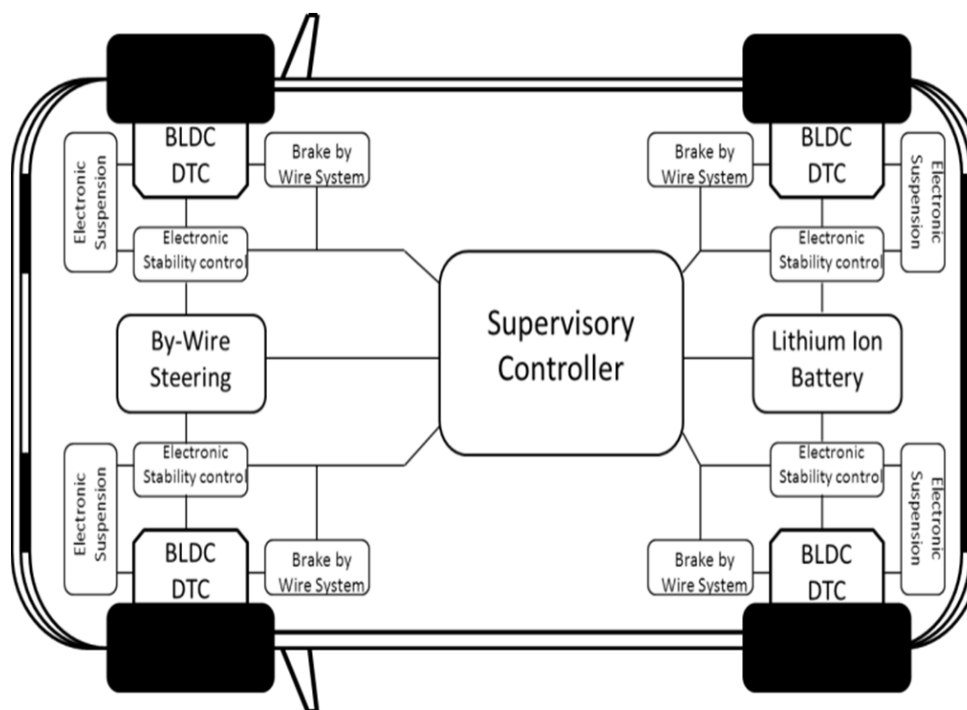


Fig.1 IFECV Drive Train System

Sensorless control methods clearly reduce the complexity of motor development, cost, and maintenance, but they also have drawbacks, such as the inability to detect back-EMF at low speeds, transient delay, and discontinuous response due to large commutation intervals. Important studies have described unique BLDC motor control algorithms [3]-[7]. For BLDC vehicles, a sensorless control approach with a novel flux-linkage feature has been proposed [3]. This technique improves the problem of sensorless control at low speeds. “G(.)” is a velocity-independent function of variables defined with consideration of the rotor mechanical viewpoint. This method can estimate the rotor position at about 1.5% of the rated speed. Since control from a stall position is required, it is suitable for in-wheel applications. For BLDC drives, a four-switch converter using a modern regulated PWM control technique has been suggested [4]. The main issues with the suggested method are the generation of 120° conducting current profiles in the three-phase winding and the current distortion in the two-phase path due to back-EMF of the silent phase.

A new Cdump converter, a DSP-controlled PWM chopper, and a power supply have been presented [5]. To maintain a consistent voltage-to-frequency ratio and enable a motor to operate with constant torque, a dual-speed and current closed-loop controller are employed. It has been reported that the switches' dv/dt rating and commutation are controlled by snubber circuits and pressurized commutation RC circuits. Simulation results indicate that the torque ripple of a BLDC motor increases because of electrical voltage spikes, which are not appropriate for in-wheel applications. To control the speed of small BLDC vehicles, a responsive fuzzy control technique using parallel distributed compensation has been implemented [6]. In the

event of parameter disturbance, the simulation results show stable speed control of the BLDC motor. Although torque smoothness and stability are crucial for high-performance EVs, they are also beneficial for other applications, such as electric motorcycles and scooters.

In particular, conduction-angle control and current-mode control are operating modes of a virtual controller for BLDC motors [7]. Because torque in a BLDC motor is directly correlated with current, current control affects motor torque control. With the help of the proposed virtual controller, the BLDC speed ripple is reduced by up to 3.4%. This strategy would be more suitable for EV applications if torque ripple reduction were also taken into account. The DTC method is a sensorless control method. It does not detect rotor position using sensors. One of the key components of an electric vehicle drivetrain is torque control. When torque ripple is reduced, power is delivered more smoothly to the wheel, thereby improving the safety and functionality of an electric vehicle. Therefore, the DTC switching approach is recommended for high-performance electric motors. This research proposes a direct torque control (DTC) switching method for BLDC motors used in EV propulsion systems. This paper explains the principle of the DTC switching method for three-phase BLDC motors used in electric vehicle applications. The simulation results of the proposed direct torque control switching strategy for BLDC motors are presented and discussed.

II. LITERATURE REVIEW

Kim and Lee [8] investigated advanced control techniques for BLDC motors in electric vehicles, focusing on in-wheel configurations. The study highlights the advantages of

BLDC motors, including efficiency, high torque, and compact size. To lessen torque ripple and enhance motor responsiveness, they suggested a modified Direct Torque Control (DTC) strategy. Improved performance under varying driving situations was validated by simulation and experimentation. Their work demonstrates the importance of precise motor control for EV efficiency and stability, emphasizing real-time performance improvement over traditional switching methods.

Singh and Sharma [9] presented a performance analysis of BLDC motors driven by a DTC approach for EV applications. They compared DTC with traditional sinusoidal and trapezoidal control techniques. The simulation results, conducted in MATLAB/Simulink, showed that DTC significantly reduced torque ripple and improved transient performance. Their study emphasized that precise torque control is crucial for EVs, especially under sudden acceleration or braking. The work supports DTC as a more efficient and reliable control method for modern electric vehicles utilizing BLDC motors.

Zhang and Chen [10] developed a detailed simulation model for BLDC motor control using intelligent algorithms such as fuzzy logic and adaptive DTC. Their study addressed limitations in traditional control schemes, such as torque ripple and delayed dynamic response. The proposed system showed improved accuracy, reduced torque oscillations, and enhanced adaptability under varying load conditions. Results validated through Simulink and prototype testing indicated the superiority of the intelligent DTC approach in optimizing the performance of in-wheel BLDC motors in EVs. The study offers valuable insights into integrating smart control into modern propulsion systems.

Patel and Bansal [11] analyzed various torque-control techniques used in electric car BLDC motors, including Field-Oriented Control (FOC), traditional DTC, and customized DTC. Their MATLAB-based study concluded that DTC offered better dynamic torque response and faster control with less complexity. Although FOC provided good steady-state performance, it required extensive computation and sensors. DTC, in contrast, ensured lower torque ripple and simpler implementation. The authors highlighted the importance of control-algorithm selection in EV propulsion efficiency, especially for in-wheel motor systems, and concluded that DTC holds superior promise in high-performance EV scenarios.

Alshamasin and Hasanien [12] focused on the development of a high-performance DTC method for BLDC motors in urban electric vehicle applications. They implemented a robust switching-table modification and voltage-vector optimization to enhance motor efficiency and minimize noise. Simulation results demonstrated a significant reduction in torque ripple and improved energy efficiency, particularly in frequent start-stop city driving conditions. The authors emphasized the suitability of DTC for smart mobility solutions, stating that the improved system not

only increases passenger comfort but also reduces battery stress, thereby enhancing overall system durability and range.

Rao and Mehta [13] conducted a simulative investigation into BLDC motor performance using DTC, aiming to improve the propulsion of compact electric vehicles. Their study explored torque accuracy, response time, and ripple effects under different load profiles. The MATLAB/Simulink model revealed that DTC provided faster torque settling time and stable operation even during harsh transients. They compared DTC with six-step commutation and sensorless FOC approaches, concluding that DTC offered a good balance of accuracy, simplicity, and robustness for in-wheel motor applications. Their results support DTC as a practical and scalable solution for commercial EV design.

Ahmed and Khan [14] proposed an enhanced Direct Torque Control (DTC) algorithm for brushless DC motors aimed at increasing efficiency in electric vehicle applications. Their model integrated an optimized hysteresis band and predictive voltage-vector selection to reduce switching losses and torque ripple. The results demonstrated significant improvements in motor performance during variable-speed operations, with reduced electromagnetic interference and better thermal stability. Their research confirmed that refining DTC logic could substantially improve in-wheel drive systems, particularly under urban traffic scenarios. The study supports the DTC-based approach as highly beneficial for smart and energy-conscious transportation solutions.

Li and Wu [15] developed and implemented a DTC-based control system for BLDC motors in electric vehicles. Their research addressed the common drawbacks of torque ripple and poor transient behavior by employing a novel switching logic integrated with voltage-vector prediction. The simulation, verified with prototype testing, showed notable improvements in energy efficiency, acceleration smoothness, and system stability. The study validated that DTC is more adaptable and energy-efficient than traditional sinusoidal and trapezoidal controls. Their work contributes to the advancement of EV motorization by offering a reliable, low-complexity control system suited for real-time automotive environments.

Verma and Sharma [16] explored the role of intelligent algorithms in reducing torque ripple in BLDC motor drives for electric vehicles. They implemented an adaptive fuzzy-based DTC technique, which dynamically adjusted the switching table based on real-time load and speed feedback. Their simulation model showed improved torque stability and reduced mechanical vibration in in-wheel motors. The research emphasized that intelligent controllers are essential for overcoming the limitations of conventional control strategies. Their study supports the integration of AI-driven control systems to enhance the reliability, comfort, and lifespan of EV propulsion units.

Yadav and Tripathi [17] examined BLDC motor Direct Torque Control (DTC), especially for the propulsion of electric vehicles. Their MATLAB/Simulink-based study demonstrated that DTC provides fast torque response and better performance compared with conventional control schemes. They addressed torque ripple, which is a major limitation in BLDC motors, by optimizing voltage-vector selection. The simulation confirmed improved controllability, reduced ripple amplitude, and enhanced efficiency, making the system more suitable for real-time EV operations. The study contributes to optimizing EV performance and supports further development of intelligent DTC-based propulsion technologies.

Kumar and Rani [18] compared various sensorless control strategies for BLDC motor drives used in electric vehicles, including back-EMF-based, Hall-sensor-based, and DTC-based approaches. They found that the DTC technique offers the best balance of performance, cost, and reliability, especially under varying load conditions. Simulated results indicated that DTC had faster response times, less torque ripple, and smoother operation. The study emphasized the practicality of sensorless DTC in compact EV platforms, particularly in reducing system complexity and enhancing battery life through efficient energy use.

Bose [19] provided a comprehensive review of power electronics and electric motor-drive systems used in green transportation. The paper highlighted the rising importance of BLDC motors in EV propulsion due to their high torque density, maintenance-free operation, and compatibility with digital control. He discussed how DTC methods outperform other control strategies in dynamic environments by minimizing torque and flux estimation errors. The review covered both theoretical and practical implementation aspects, emphasizing that advanced DTC-based drives can significantly improve vehicle acceleration, efficiency, and control under real-world driving conditions.

A. Research Gap

Despite significant advancements in electric vehicle (EV) technology and motor control systems, several research gaps persist in the field of BLDC motorization using Direct Torque Control (DTC). While BLDC motors have become the preferred choice for EV propulsion due to their high efficiency, torque density, and compact size, their performance is still limited by control challenges, particularly torque ripple, switching losses, and sensitivity to parameter variations. Most existing studies focus on conventional DTC or modified algorithms, yet they often fail to address real-time adaptability, optimal voltage-vector selection, and ripple minimization simultaneously. Additionally, limited research has been conducted on intelligent or hybrid DTC strategies that integrate AI techniques such as fuzzy logic or neural networks to enhance performance under dynamic load conditions.

Moreover, many simulation models in prior research lack validation through experimental data or fail to account for in-wheel motor applications, which involve additional complexities such as uneven load distribution and wheel dynamics. The absence of robust, experimentally validated, intelligent DTC algorithms tailored for in-wheel EV systems presents a critical gap. Addressing this gap is essential to improve system reliability, energy efficiency, and driving comfort, ultimately supporting the widespread adoption of EVs in both urban and long-range transportation applications.

III. METHODOLOGY

A. BLDC Motor Design and Engineering

A BLDC motor consists of two main structural components: the rotor (spinning part) and the stator (stationary part). These components create a magnetic field that rotates at the same frequency, thereby preventing slip as in induction machines. Permanent magnets are mounted on the rotor, and armature windings are placed on the stator.

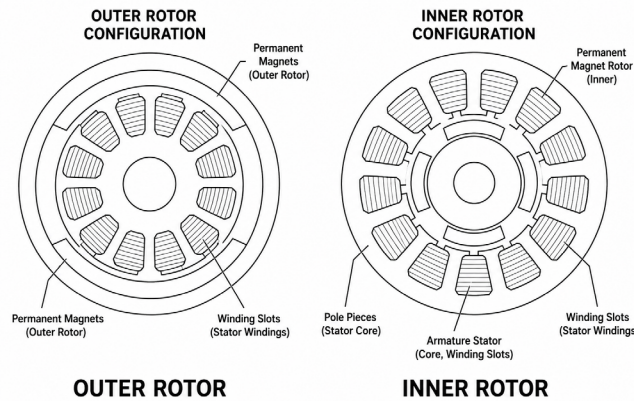


Fig.2 Outer and Inner Rotor

The rotor surface of BLDC motors is equipped with permanent magnets that provide continuous excitation. As

illustrated in Figure 2, there are two main rotor designs: outer rotor and inner rotor. In the outer-rotor layout, the

windings are placed in the middle of the motor. By acting as an insulator and surrounding the stator windings, the rotor magnets reduce the motor's heat dissipation capability. This design is used in low-speed applications and operates at lower rated current or duty cycles. One advantage of this design is very low cogging torque. In the inner-rotor arrangement, the stator winding is fixed to the motor housing and surrounds the rotor. The main advantages of this arrangement are better heat dissipation, greater power capability, and lower friction; therefore, it is used in high-speed applications.

The number of stator slots is determined by the rotor poles, winding configuration, and number of phases. The stator comprises three-phase windings. It is better to use a

fractional slot/pole arrangement to reduce cogging torque. The stator has winding slots, and the change in current polarity within the slots supplies power to the rotating magnetic field. Rotor position is required in order to change current polarity according to the rotor magnetic field. This information is continuously provided by Hall-effect sensors on the stator. For current commutation, solid-state switches are used, eliminating the need for brushes.

B. Standard Control System

One simple way to satisfy these requirements is to use current-control techniques. The construction of the current controller for a BLDC motor is shown in Figure 3. Three-phase current can be controlled at its source.

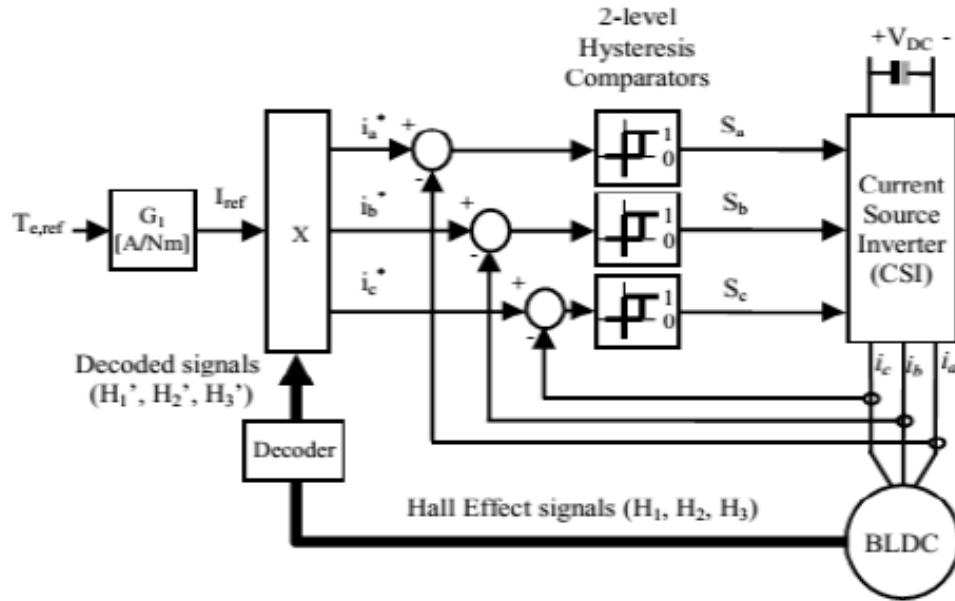


Fig.3 Traditional Current Control Controller for a BLDC Motor

The motor currents need to be changed in line with their references (i_a^* , i_b^* , and i_c^*), as shown in Figure 3. The torque demand (i.e., $I_{ref} = T_{e,ref} \times G1$) and the decoded signals ($H1'$, $H2'$, and $H3'$) generated from the Hall-effect data ($H1$, $H2$, and $H3$) form the basis for reference current generation, as shown in Table I.

TABLE I DECODED SIGNALS DERIVED FROM HALL SIGNALS

Hall effect signal			Decoded signal		
H1	H2	H3	H1	H2	H3
0	0	0	0	0	0
0	0	1	0	-1	+1
0	1	0	-1	+1	0
0	1	1	-1	0	+1
1	0	0	+1	-1	0
1	0	1	0	+1	-1
1	1	0	0	+1	-1
1	1	1	0	0	0

A two-level hysteresis comparator is used to control each phase current. Its job is to supply the inverter with the proper switching state, either to increase or decrease the

phase current, so that the error (or current ripple) is kept within the hysteresis band (HB). In this way, each phase reference current has the same waveform pattern as the corresponding decoded signals.

Even when the reference current is set to zero, current is still drawn from the batteries using the conventional current-control method. When the accelerator pedal is released ($T_{e,ref} = 0$), the actual motor torque is reduced to zero, but the phase current still needs to be controlled by the IGBTs/MOSFETs within the inverter. For electric vehicle applications, a current-blocking technique is therefore required to prevent waste of battery power when $T_{e,ref} = 0$.

The use of a hysteresis comparator, as demonstrated in Figure 4, within the hysteresis band at about 0 A can be implemented with only a slight alteration to the original current-control design shown in Figure 3. The proposed current-blocking method turns everything off to prevent current from being drawn from the battery. According to

Figure 4, activation of the current-blocking method requires the absolute value of torque, $T_{e,ref}$, and phase currents

(i_a , i_b , and i_c), which are then applied to a hysteresis comparator and a zero-crossing detector, respectively.

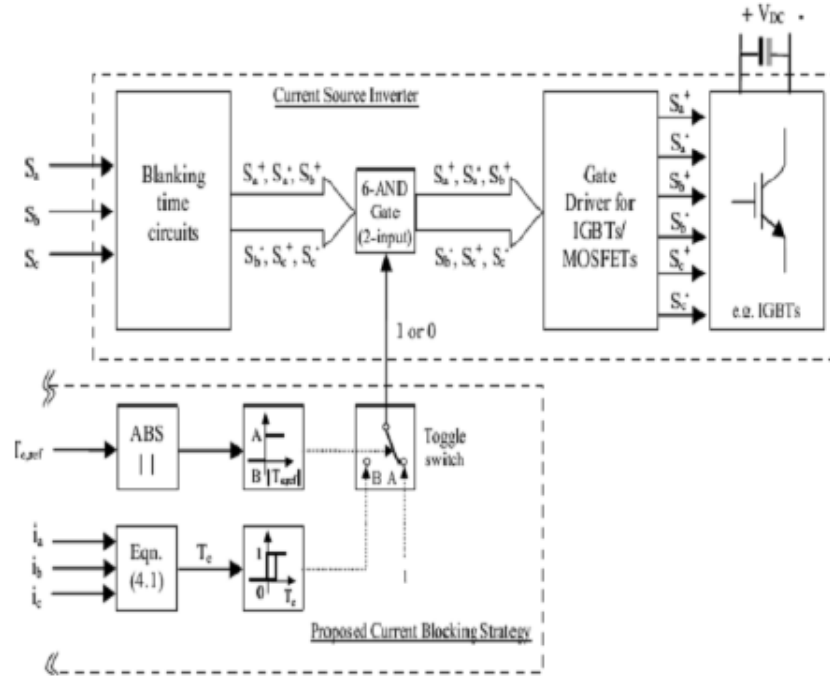


Fig.4 Conventional Current-Blocking Technique Using a Hysteresis Comparator

If the torque production, T_e , and torque demand, $T_{e,ref}$, drop to zero, the current-source inverter turns off all IGBTs and MOSFETs. Figure 5 provides a clearer illustration of

the activation condition. Otherwise, standard switching operations are carried out to keep the torque (or current) within the hysteresis band.

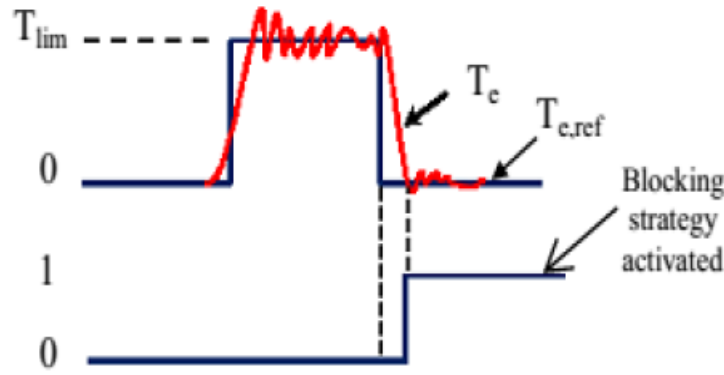


Fig.5 Activation of the Blocking Strategy When T_e and $T_{e,ref}$ Decline

IV. CONCLUSION

In order to raise the voltage level for batteries used in electric vehicles between recharges, this study presents simulation results for direct torque control driving of brushless DC (BLDC) motors using the present hysteresis-control approach. Under regular current control, the batteries continue to discharge current even when the motor is off. A zero-crossing sensor is installed to stop that power leakage. A hysteresis comparator creates a high switching frequency, which results in torque ripple. However, only current control is possible in this process. By employing the automatic torque controller to regulate high-frequency switching and inverter switching, the proposed technique

reduces torque ripple. Direct torque control simulation is carried out in MATLAB/Simulink.

Declaration of Conflicting Interests

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Use of Artificial Intelligence (AI)-Assisted Technology for Manuscript Preparation

The authors confirm that no AI-assisted technologies were used in the preparation or writing of the manuscript, and no images were altered using AI.

REFERENCES

- [1] F. N. Azam, A. Jidin, N. A. Ngatiman, M. H. Jopri, M. Manap, A. L. Herlino, and N. F. Alias, "Current control of BLDC drive for EV application," in *Proc. IEEE 7th Int. Power Engineering and Optimization Conf. (PEOCO)*, Langkawi, Malaysia, Jun. 2013, pp. 240–244.
- [2] P. Lefley, L. Petkovska, and G. Cvetkovski, "Optimization of the design parameters of an asymmetric brushless DC motor for cogging torque minimization," in *Proc. 14th Eur. Conf. Power Electronics and Applications (EPE)*, Birmingham, UK, 2011, pp. 1–10.
- [3] N. Bahari, A. Jidin, A. R. Abdullah, and M. N. Othman, "Modeling and simulation of torque hysteresis controller for brushless DC motor drives," in *IEEE Symp. Industrial Electronics and Applications (ISIEA)*, Bandung, Indonesia, Sep. 2012, pp. 152–155.
- [4] J. S. Mayer and O. Wasynczuk, "Analysis and modelling of a single-phase brushless DC motor drive system," *IEEE Trans. Energy Convers.*, vol. 4, no. 3, pp. 473–479, Sep. 1989.
- [5] A. Jidin, N. R. N. Idris, A. H. M. Yatim, T. Sutikno, and M. E. Elbuluk, "An optimized switching strategy for quick dynamic torque control in DTC-hysteresis-based induction machines," *IEEE Trans. Ind. Electron.*, vol. 58, no. 8, pp. 3391–3400, Aug. 2011.
- [6] N. B. Bahari, A. B. Jidin, A. R. B. Abdullah, M. N. B. Othman, and M. B. Manap, "Modeling and simulation of torque hysteresis controller for brushless DC motor drives," in *IEEE Symp. Industrial Electronics and Applications (ISIEA)*, 2012, pp. 152–155.
- [7] J. R. Hendershot and T. Miller, *Design of Brushless Permanent Magnet Motors*, Oxford, UK: Magna Physics Publications & Oxford University Press, 1994.
- [8] H. Kim and S. Lee, "Advanced direct torque control of BLDC motors for electric vehicles in in-wheel drive applications," *IEEE Trans. Veh. Technol.*, vol. 66, no. 9, pp. 7780–7790, Sep. 2017.
- [9] R. Singh and A. Sharma, "Performance analysis of BLDC motors driven by a DTC approach for EV applications," *IEEE Access*, vol. 8, pp. 112450–112461, 2020.
- [10] Y. Zhang and H. Chen, "Simulation model and intelligent direct torque control of in-wheel BLDC motor drives for electric vehicles," *IEEE Trans. Transp. Electrification*, vol. 8, no. 2, pp. 2105–2116, Jun. 2022.
- [11] M. Patel and R. Bansal, "Comparative evaluation of torque-control techniques for electric vehicle propulsion using BLDC motors," *IEEE Trans. Power Electron.*, vol. 33, no. 11, pp. 9500–9512, Nov. 2018.
- [12] M. Alshamasin and H. M. Hasanien, "High-performance direct torque control scheme for BLDC motors in urban electric vehicles," *IET Electr. Power Appl.*, vol. 15, no. 4, pp. 485–496, Apr. 2021.
- [13] K. S. Rao and V. Mehta, "Investigation into dynamic response and torque ripple reduction of DTC-based BLDC drive for compact electric vehicles," *J. Electr. Eng. Technol.*, vol. 14, no. 5, pp. 1985–1995, Sep. 2019.
- [14] M. Ahmed and M. A. Khan, "Enhanced direct torque control algorithm for brushless DC motor drives in electric vehicle applications," *IEEE Trans. Energy Convers.*, vol. 35, no. 3, pp. 1420–1430, Sep. 2020.
- [15] X. Li and J. Wu, "Predictive direct torque control system for in-wheel BLDC motors in electric vehicles," *IEEE Trans. Ind. Inform.*, vol. 19, no. 5, pp. 6210–6220, May 2023.
- [16] P. Verma and S. Sharma, "Adaptive fuzzy-based direct torque control for torque ripple minimization in BLDC motor drives," *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 8, pp. 2250–2260, Aug. 2021.
- [17] A. Yadav and S. Tripathi, "Direct torque control and ripple mitigation of BLDC motor drive for electric vehicle propulsion," *Int. J. Electr. Power Energy Syst.*, vol. 108, pp. 312–322, Jun. 2019.
- [18] Kumar and U. Rani, "Comparative study of sensorless control strategies for BLDC motor drives in electric vehicles," *IEEE Power Energy Technol. Syst. J.*, vol. 5, no. 4, pp. 145–154, Dec. 2018.
- [19] K. Bose, "Power electronics and motor drive systems in electric and hybrid vehicles for green transportation: A review," *Proc. IEEE*, vol. 105, no. 11, pp. 2152–2170, Nov. 2017.